Assignment 3

1. Is it possible to create a hard link to a directory in Linux? If not, why?

No, normally it is not possible to create a hard link to a directory in Linux.A hard link points directly to the inode of a file. If hard links to directories were allowed, it could create circular directory structures, making it difficult for the file system to maintain the directory hierarchy and perform operations such as traversal and deletion safely.For this reason, Linux allows hard links to regular files but does not allow ordinary users to create hard links to directories.

2. Is it possible to create a file or directory in Linux with a name that contains a space?

Yes. Linux allows spaces in file and directory names.

However, because the shell normally treats a space as a separator between arguments, the name must be enclosed in quotes or the space must be escaped using \.

For example, to create a directory named My Documents:

Using quotes - `mkdir "My Documents"`

Using an escape character - `mkdir My\ Documents`

Both commands create a directory with the name : `My Documents`

3. Can we create a file and a directory with the same name in the current working directory?

No, we cannot create both a file and a directory with exactly the same name in the same directory.

For example, first create a file named sample : `touch sample`

Now try to create a directory with the same name : `mkdir sample`

The command produces an error similar to:

```
mkdir: cannot create directory 'sample': File exists
```

Similarly, if we first create the directory : `mkdir sample`

and then try : `touch sample`

the command cannot create a separate file with that name because an entry named sample already exists.In a single directory, each name must uniquely identify one

directory entry. The name sample cannot refer to both a file and a directory at the same time.sample can be either a file or a directory, but not both simultaneously.

4. What is the ls command (with options) equivalent to the dir command?

In Linux, the `ls` command is used to list files and directories, similar to the `dir` command in Windows.

Windows	**Linux**	**Purpose**
dir	ls	List files and directories
dir /w	ls	List files in a compact format
dir /a	ls -a	Show hidden files
dir with detailed information	ls -l	Show detailed file information
dir /a + details	ls -la	Show hidden files with detailed information